

Motor Control Board

Variant: DRAFT

2026-08-21

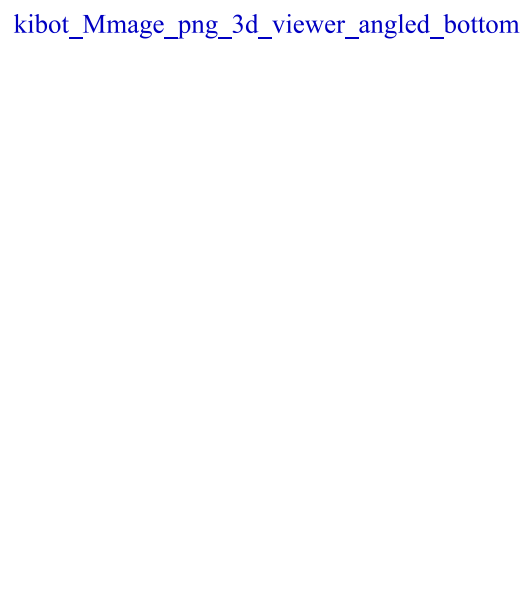
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

NOTES

Comment

Not fitted components are marked as **X**

- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

Date: 21-Aug-2026

Block Diagram

File: Block Diagram.kicad_sch

Revision History

File: Revision History.kicad_sch

Project Architecture

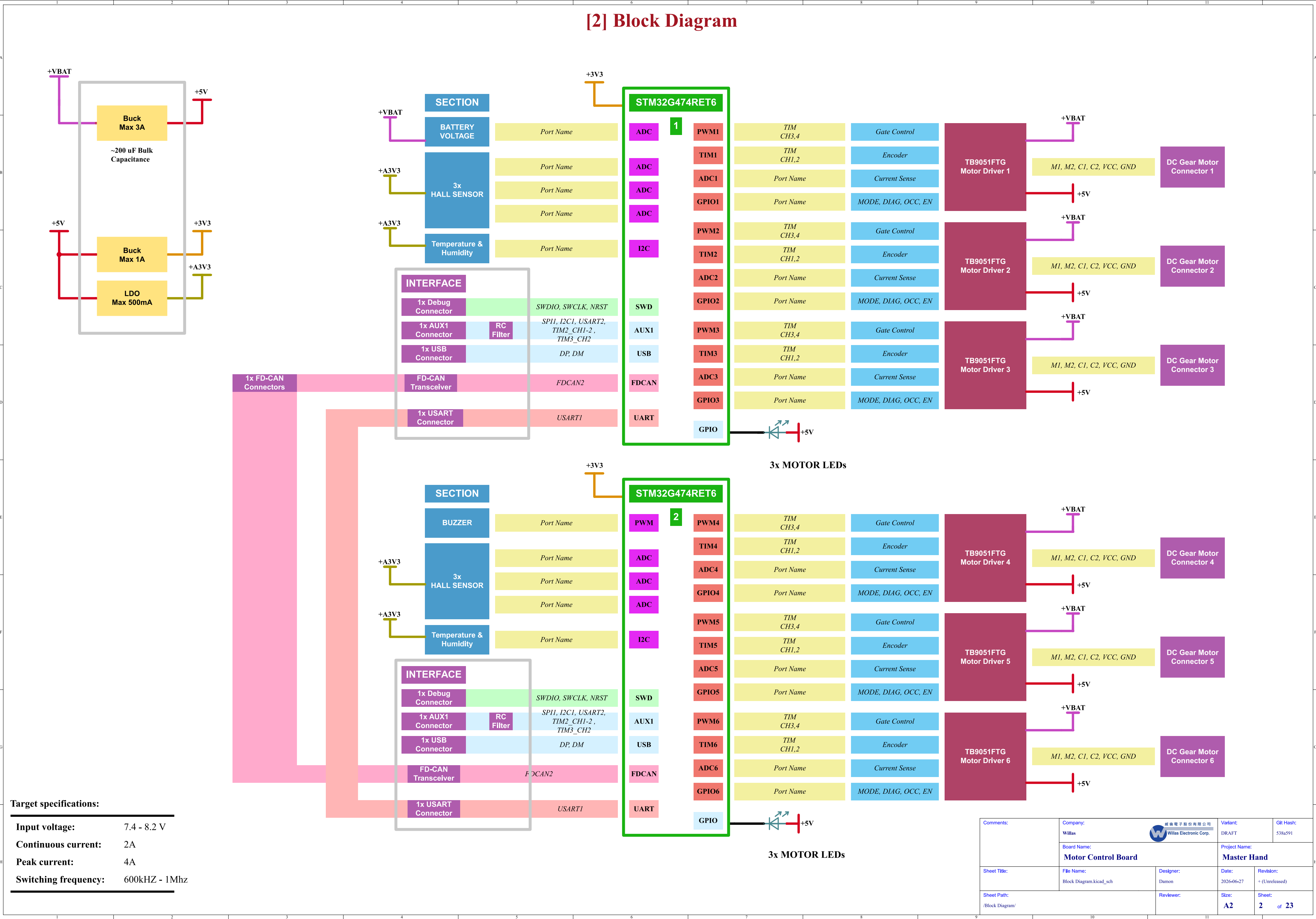
File: Project Architecture.kicad_sch

Power - Sequencing

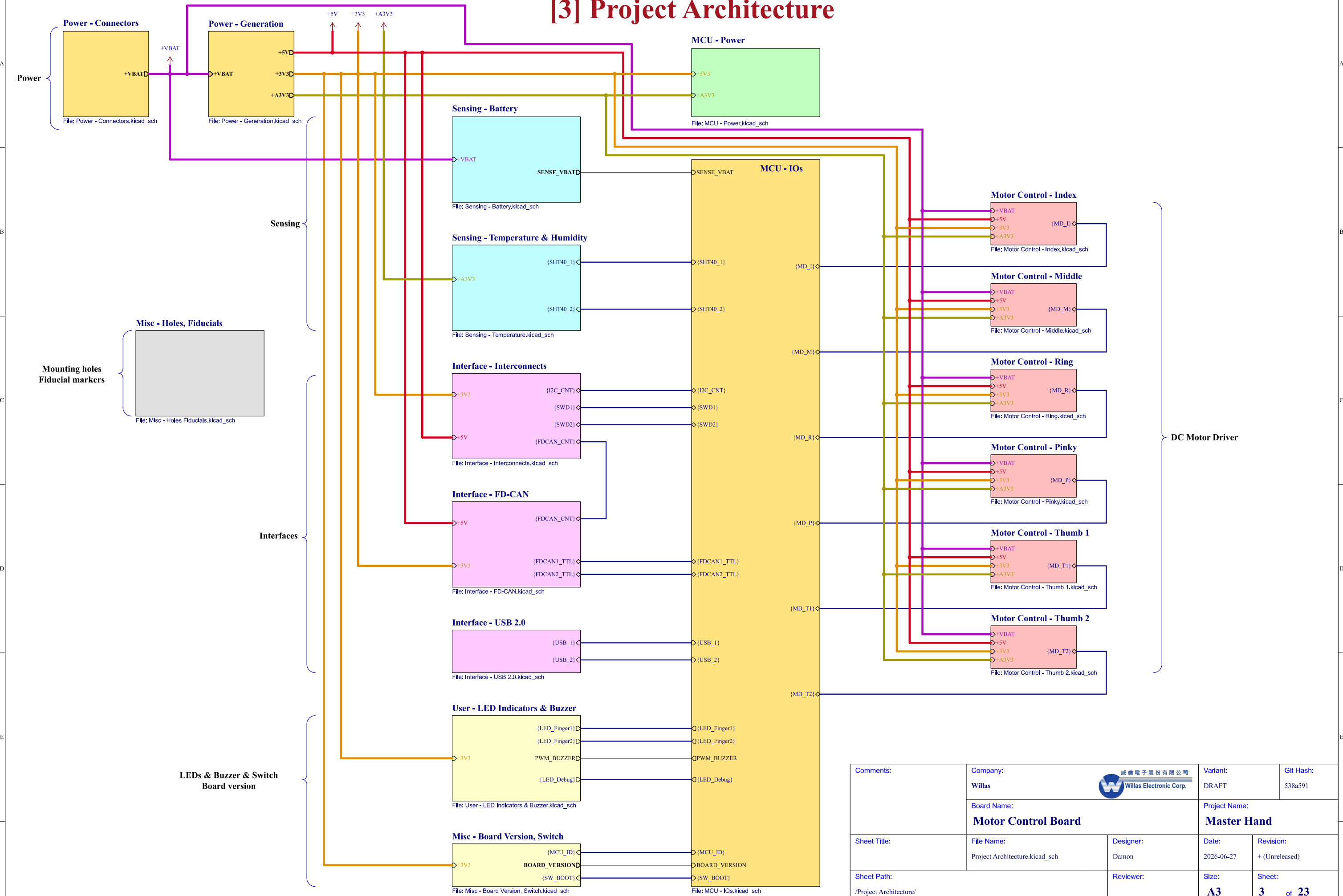
File: Power - Sequencing.kicad_sch

Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: 538a591	
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Master_Hand_Kicad.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 23	

[2] Block Diagram

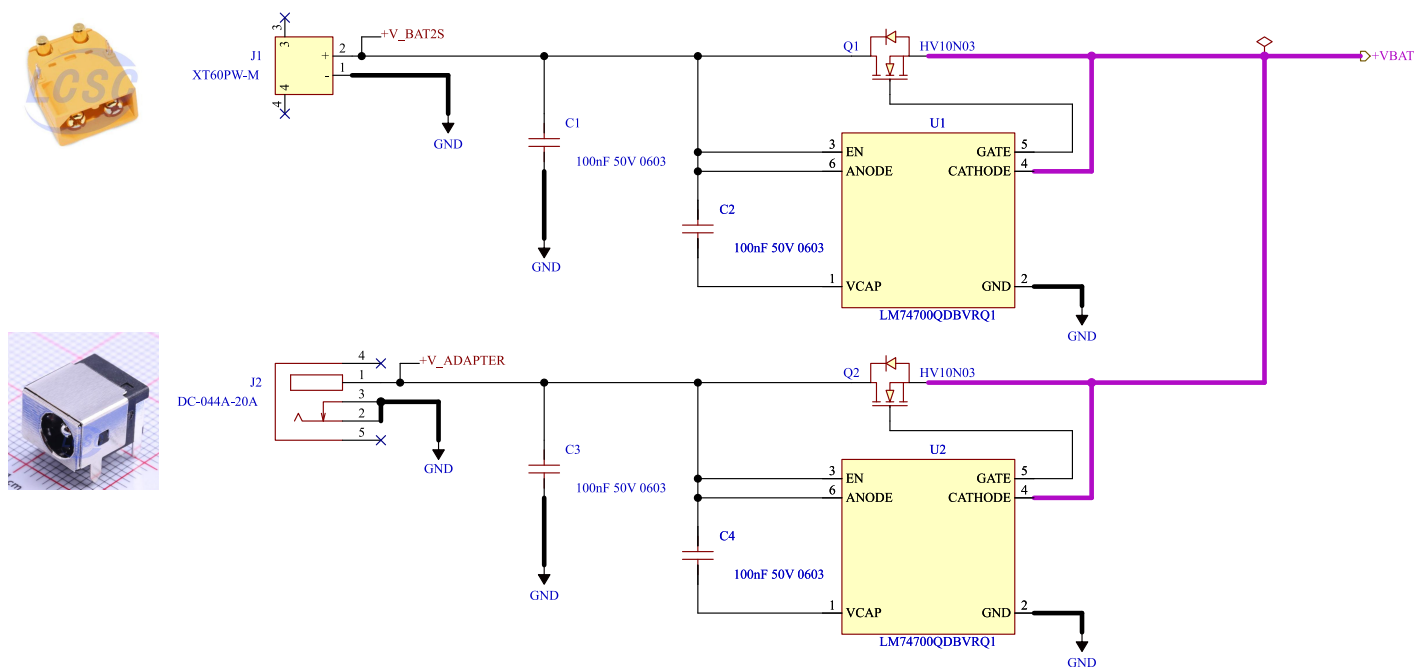


[3] Project Architecture



Comments:	Company: Willas 		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Project Architecture.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/	Reviewer:	Size: A3	Sheet: 3 of 23	

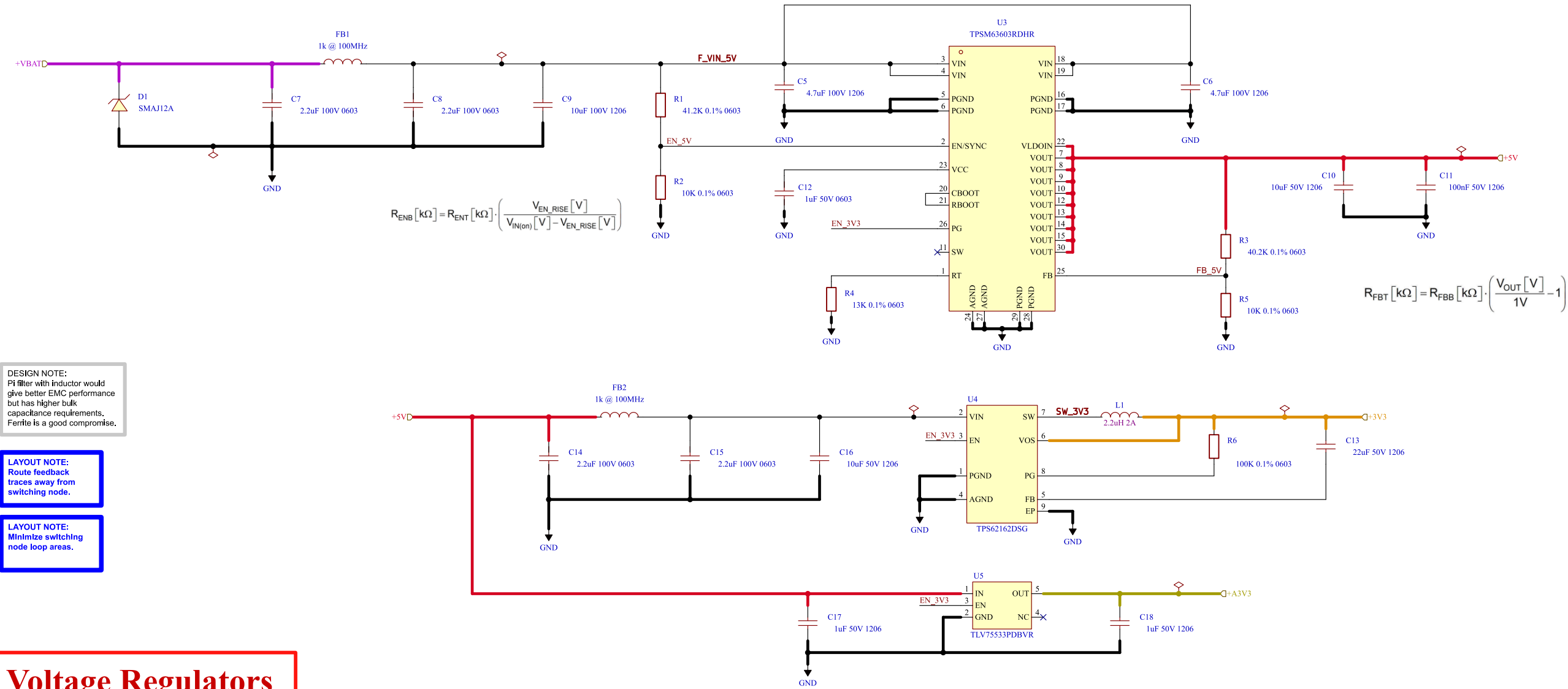
[4] Power - Connectors



Power Connectors - Path Control

Comments:	Company: Willas		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Power - Connectors.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 4 of 23

[5] Power - Generation



Voltage Regulators

DC BULK

DC-Link Bulk Capacitors. 330uF + 10 x 10 uF = 430uF (35V)

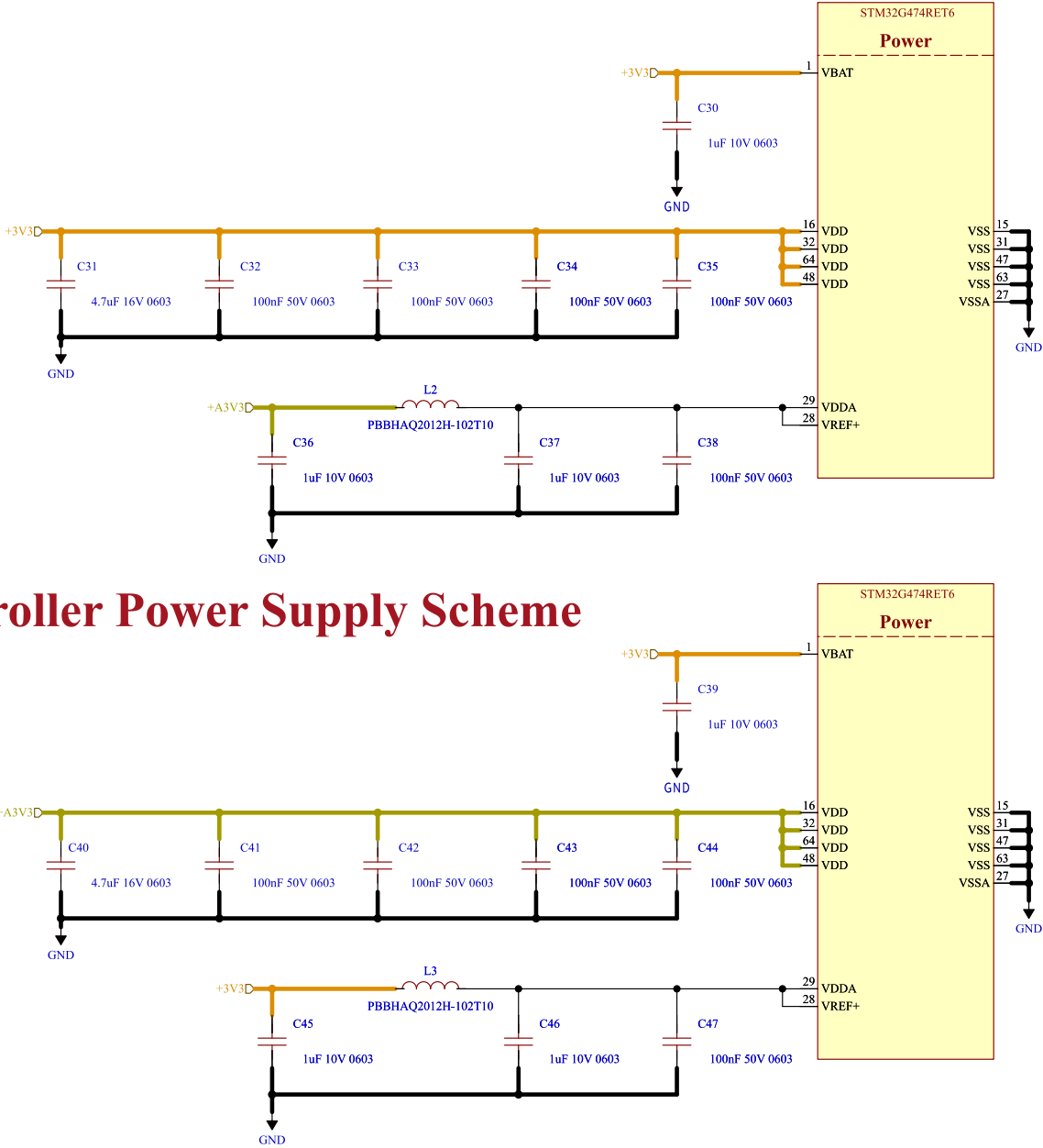
Test Points



Comments:	Company: Willas		 Willas Electronic Corp.	Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Power - Generation.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Power - Generation/	Reviewer:		Size: A3	Sheet: 5 of 23	

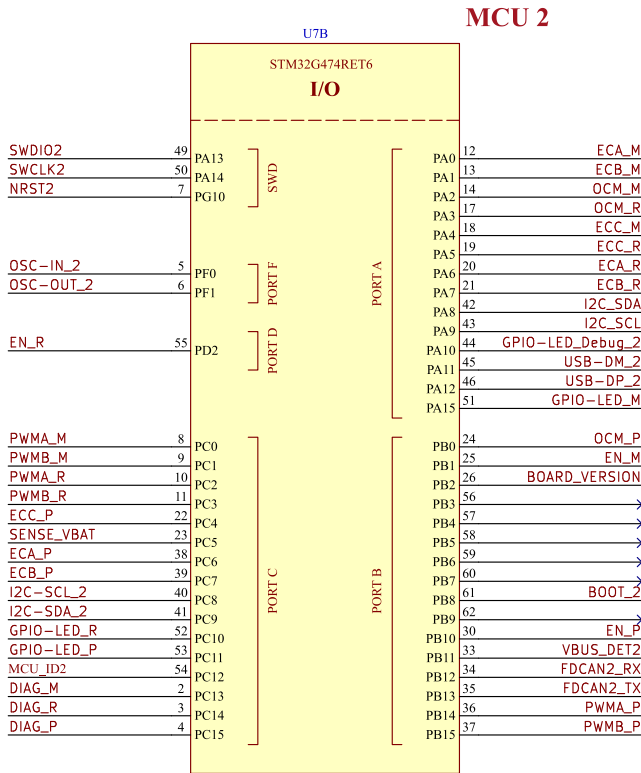
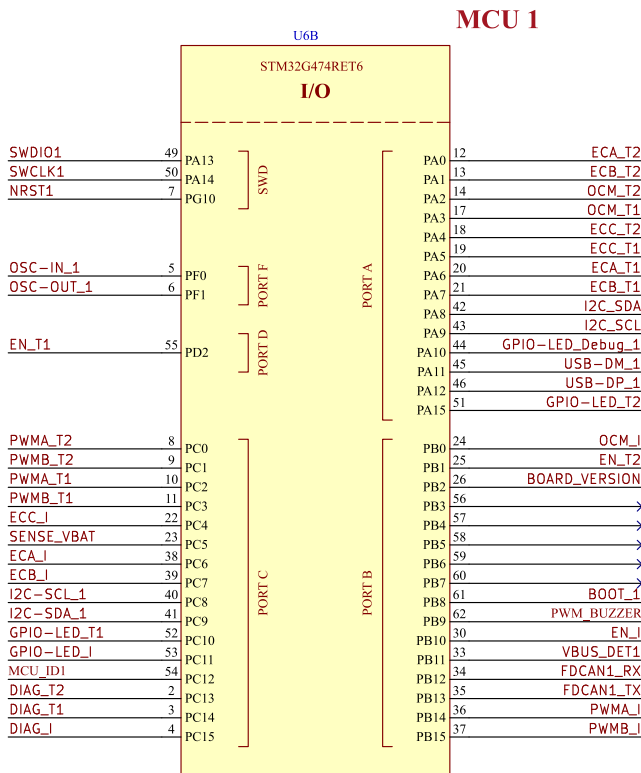
Microcontroller Power Supply Scheme

LAYOUT NOTE:
Place each capacitor close
to the relevant pin.
Minimize loop inductance.



Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: MCU - Power.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/MCU - Power/	Reviewer:		Size: A3	Sheet: 6 of 23		

[7] MCU - IOs



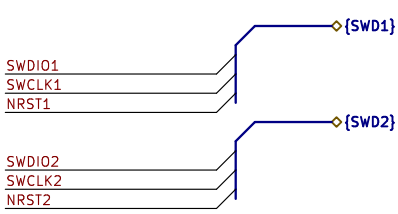
DESIGN NOTE:
We use the LQFP64 version to have extra pins for controlling an external fan.

DESIGN NOTE:
Do not forget to flash option bytes when flashing for the first time.

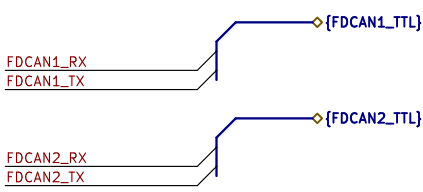
LAYOUT NOTE:
Footprint deviates from datasheet and IPC7351 to allow for soldermask sliver between pads according to PCBWay design rules.

SENSE_VBAT → SENSE_VBAT

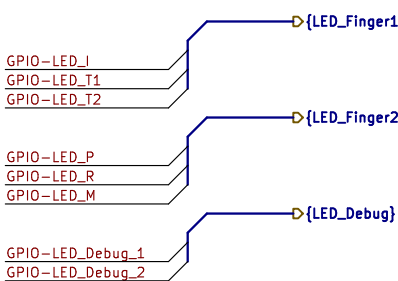
Battery Voltage Sense



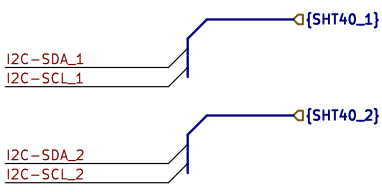
SWD (Debug) Port



5Mbps FD-CAN Bus



LEDs



Temperature and Humidity



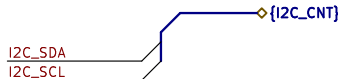
USB PORT

BOARD_VERSION → BOARD_VERSION

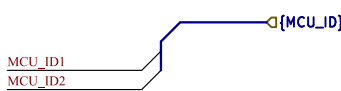
Board Version Selection

PWM_BUZZER → PWM_BUZZER

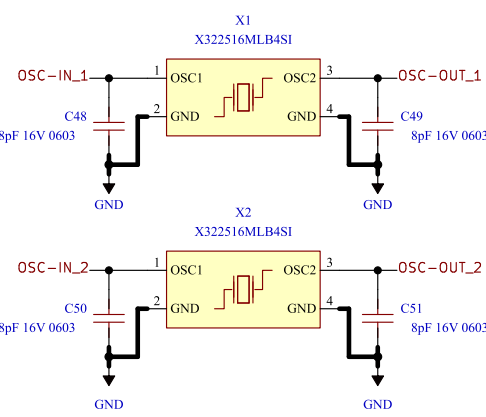
Buzzer



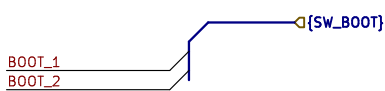
MCB to MB - I2C



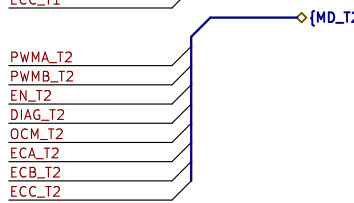
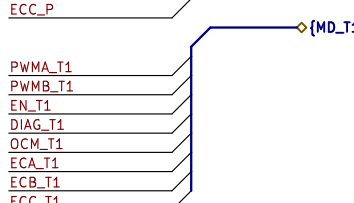
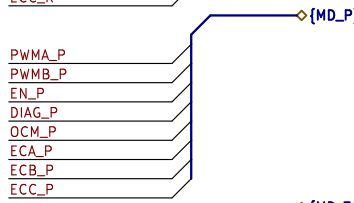
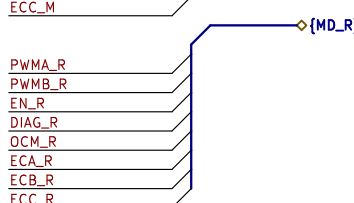
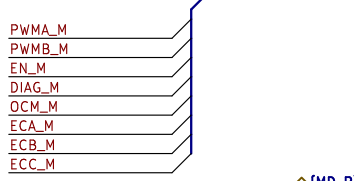
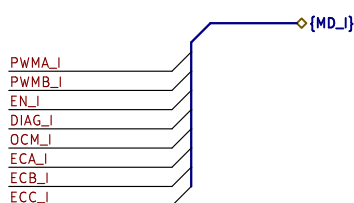
MCU ID



Oscillator HSE



SWITCH



Motor Driver

Battery Voltage Sense

SENSE_VBAT → TP9

Board Version Selection

BOARD_VERSION → TP13

SWD (Debug) Port

NRST2 → TP17
SWCLK2 → TP19
SWDIO2 → TP21
NRST1 → TP22
SWCLK1 → TP24
SWDIO1 → TP26

5Mbps FD-CAN Bus

FDCAN1_TX → TP29
FDCAN1_RX → TP31
FDCAN2_TX → TP34
FDCAN2_RX → TP36

LEDs

GPIO-LED_I → TP38
GPIO-LED_T1 → TP40
GPIO-LED_T2 → TP42
GPIO-LED_P → TP44
GPIO-LED_R → TP46
GPIO-LED_M → TP48
GPIO-LED_Debug_1 → TP50
GPIO-LED_Debug_2 → TP52

Temperature and Humidity

I2C-SDA_1 → TP54
I2C-SCL_1 → TP56
I2C-SDA_2 → TP58
I2C-SCL_2 → TP60

USB PORT

USB-DP_1 → TP64
USB-DM_1 → TP66
VBUS_DET1 → TP67
USB-DP_2 → TP69
USB-DM_2 → TP71
VBUS_DET2 → TP73

MCB to MB - I2C

I2C_SCL → TP77
I2C_SDA → TP79

Buzzer

PWM_BUZZER → TP81

MCU ID

MCU_ID1 → TP86
MCU_ID2 → TP88

MD Index

PWMA_I → TP10
PWMB_I → TP11
EN_I → TP12
DIAG_I → TP14
OCM_I → TP15
ECA_I → TP16
ECB_I → TP18
ECC_I → TP20

MD Middle

PWMA_M → TP23
PWMB_M → TP25
EN_M → TP27
DIAG_M → TP28
OCM_M → TP30
ECA_M → TP32
ECB_M → TP33
ECC_M → TP35

MD Ring

PWMA_R → TP37
PWMB_R → TP39
EN_R → TP41
DIAG_R → TP43
OCM_R → TP45
ECA_R → TP47
ECB_R → TP49
ECC_R → TP51

MD Pinky

PWMA_P → TP53
PWMB_P → TP55
EN_P → TP57
DIAG_P → TP59
OCM_P → TP61
ECA_P → TP62
ECB_P → TP63
ECC_P → TP65

MD Thumb 1

PWMA_T1 → TP68
PWMB_T1 → TP70
EN_T1 → TP72
DIAG_T1 → TP74
OCM_T1 → TP75
ECA_T1 → TP76
ECB_T1 → TP78
ECC_T1 → TP80

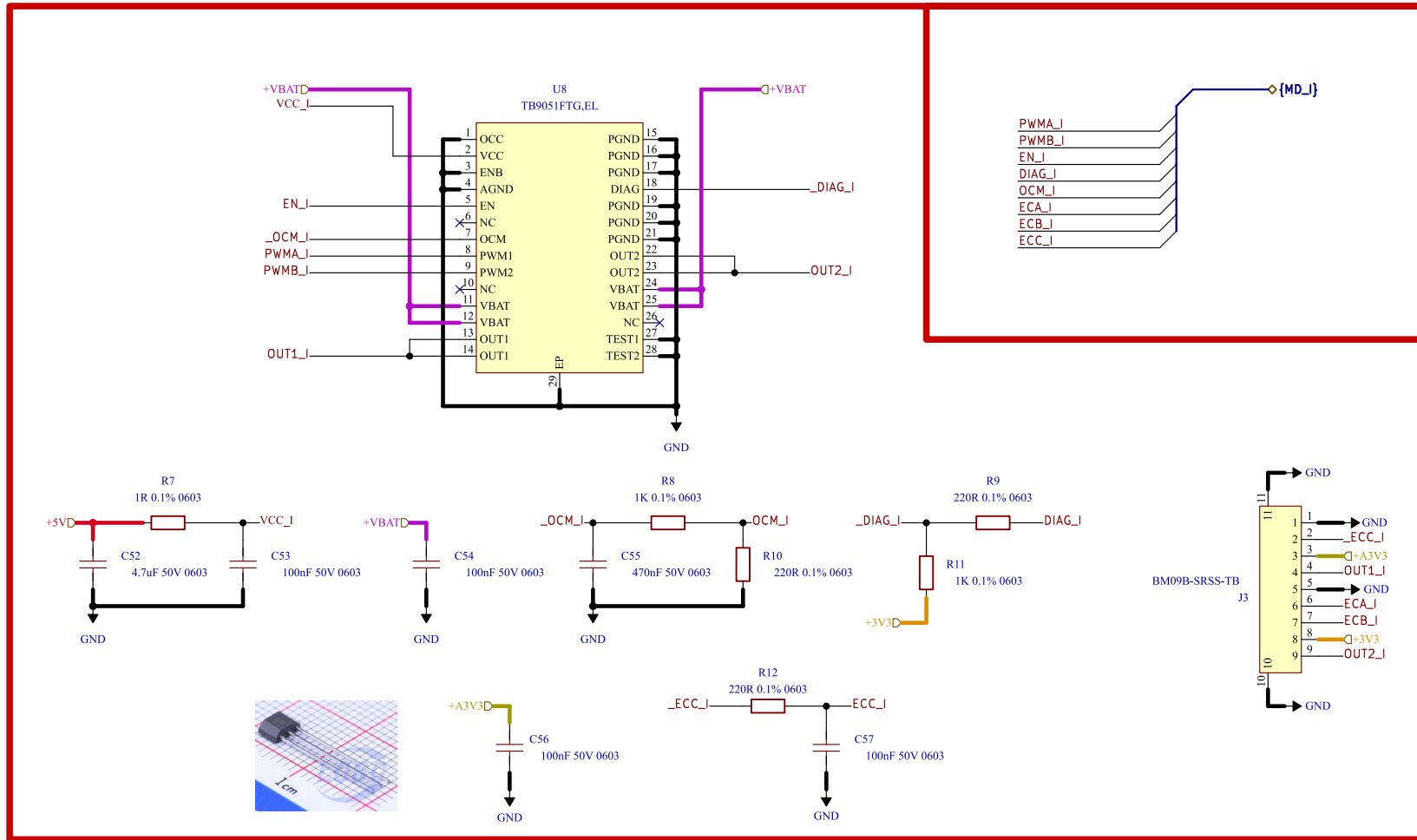
MD Thumb 2

PWMA_T2 → TP82
PWMB_T2 → TP83
EN_T2 → TP84
DIAG_T2 → TP85
OCM_T2 → TP87
ECA_T2 → TP89
ECB_T2 → TP90
ECC_T2 → TP91

Test Points

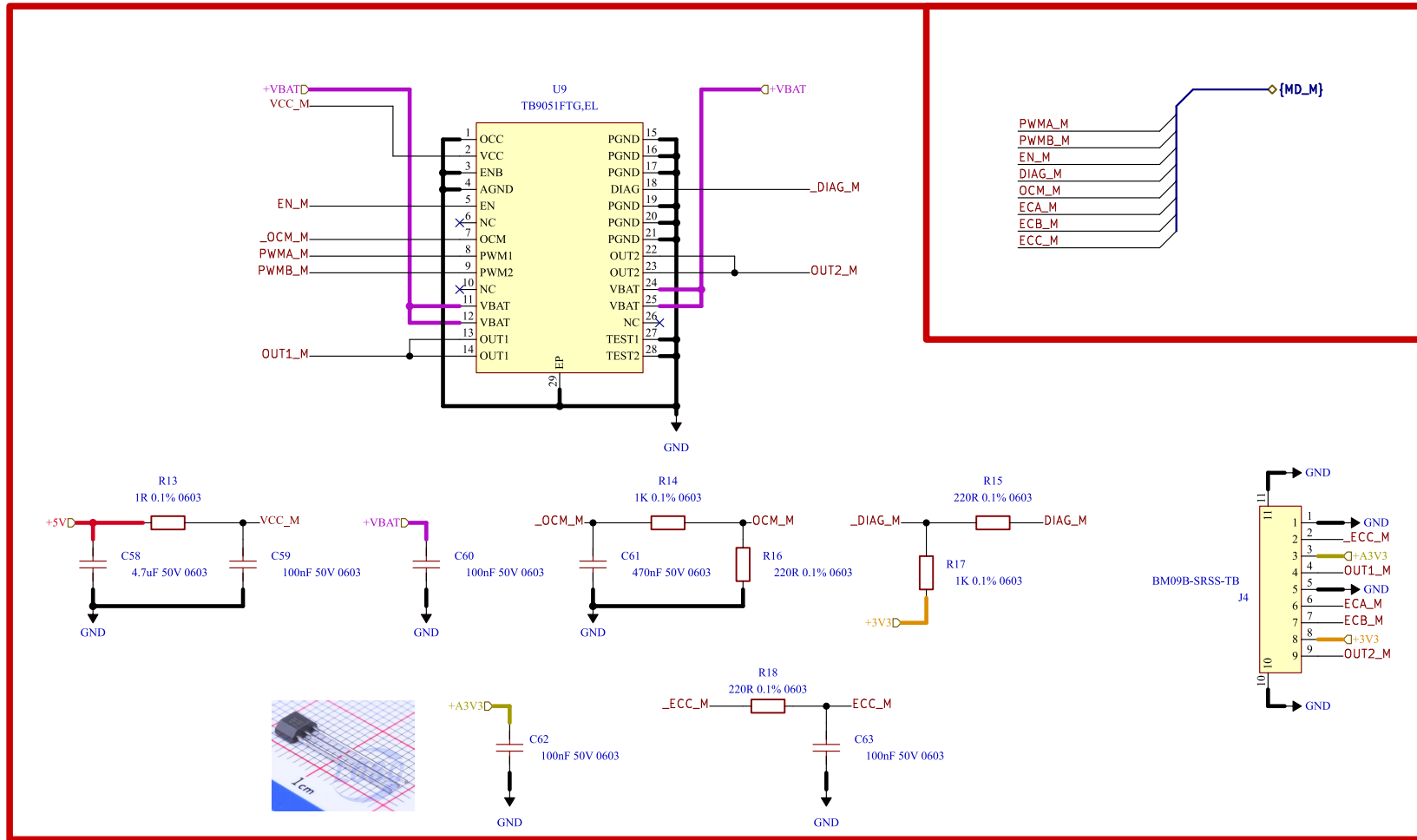
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Sheet Title:	Board Name: Motor Control Board	Project Name: Master Hand	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - IOs/	File Name: MCU - IOs.kicad_sch	Designer: Damon	Size: A3
		Reviewer:	Sheet: 7 of 23

[8] Motor Control - Index



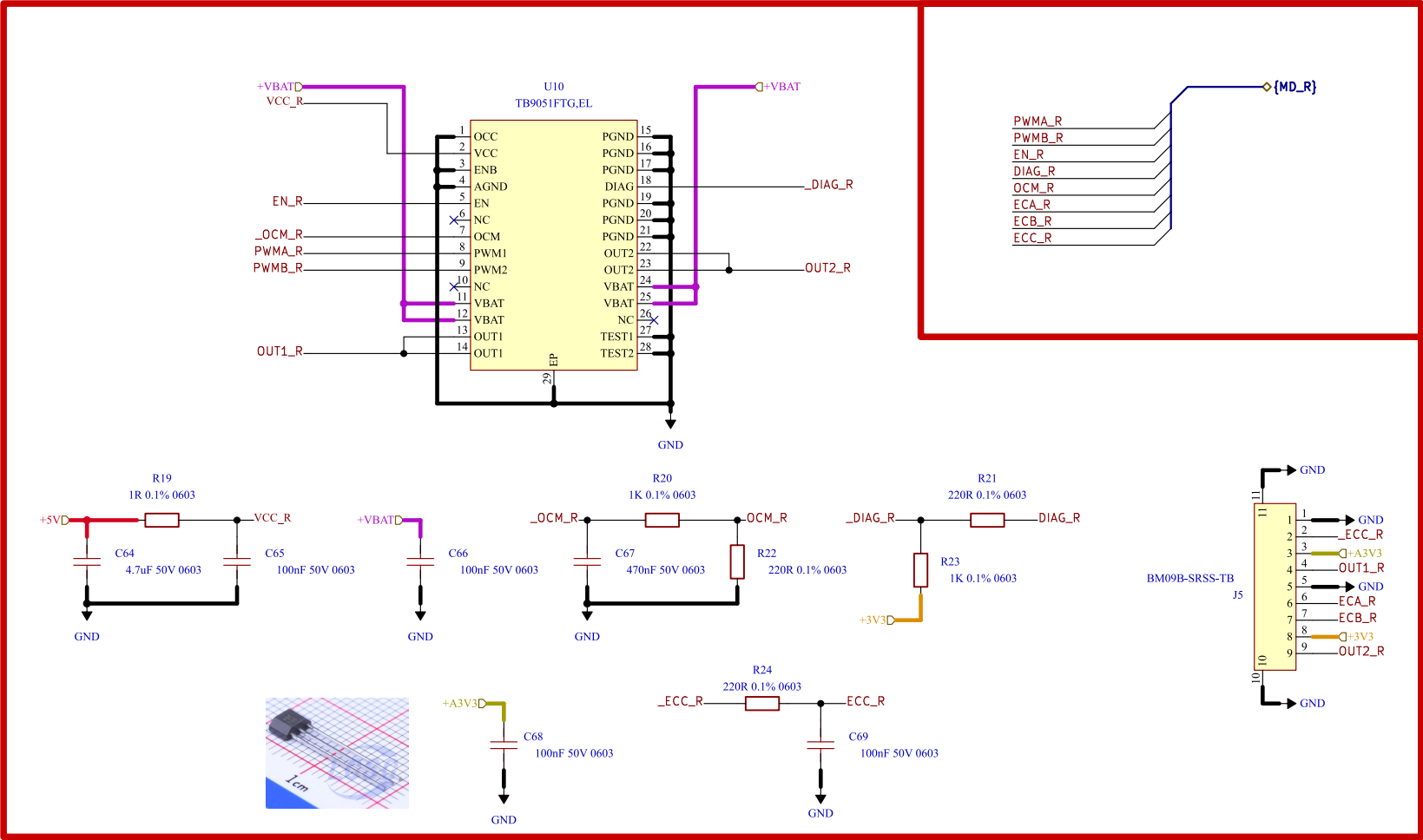
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Sheet Title:	File Name: Motor Control - Index.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Index/		Reviewer:	Size: A4	Sheet: 8 of 23

[9] Motor Control - Middle



Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Middle.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Middle/		Reviewer:	Size: A4	Sheet: 9 of 23

[10] Motor Control - Ring



Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

538a591

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Motor Control - Ring.kicad_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Motor Control - Ring/

Reviewer:

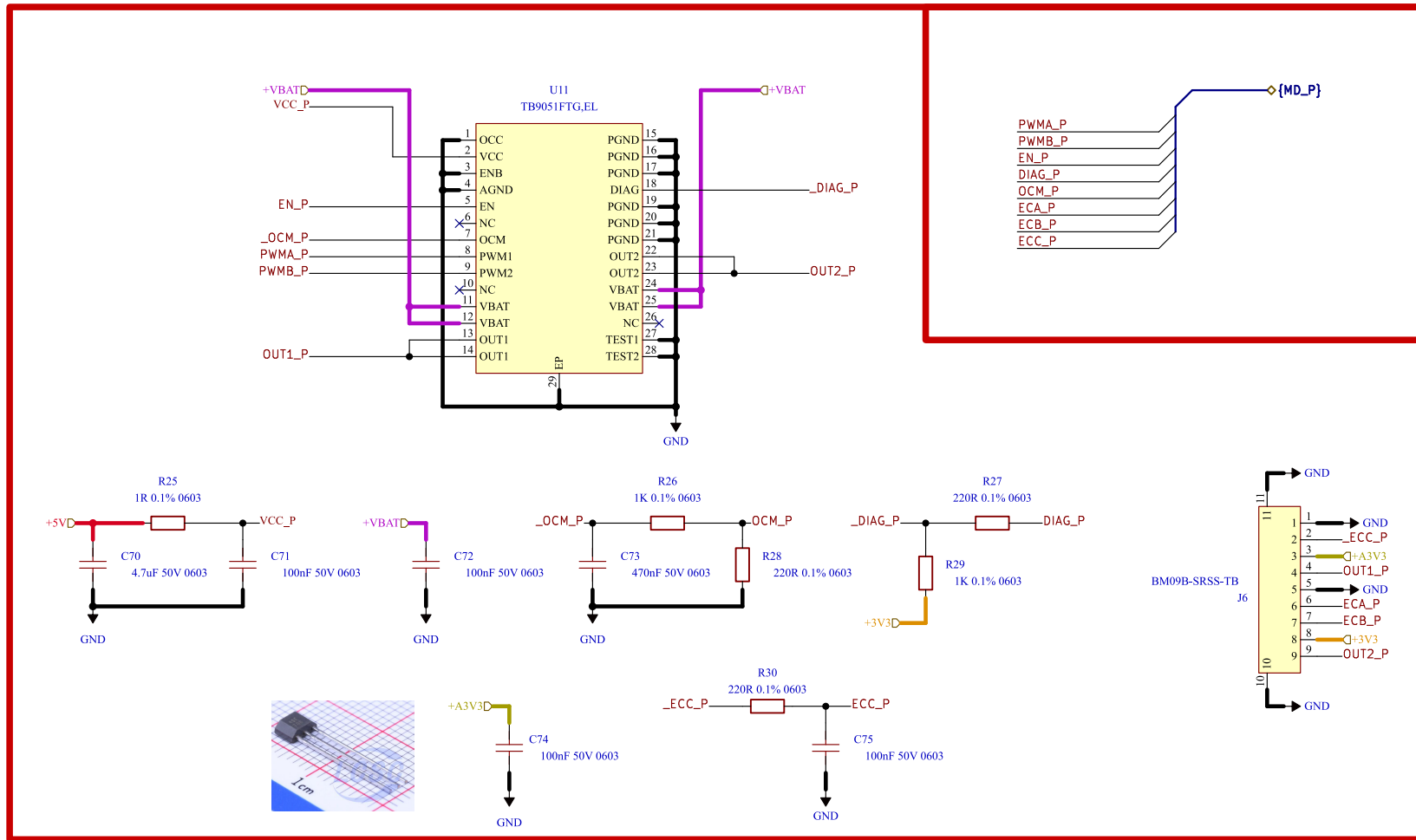
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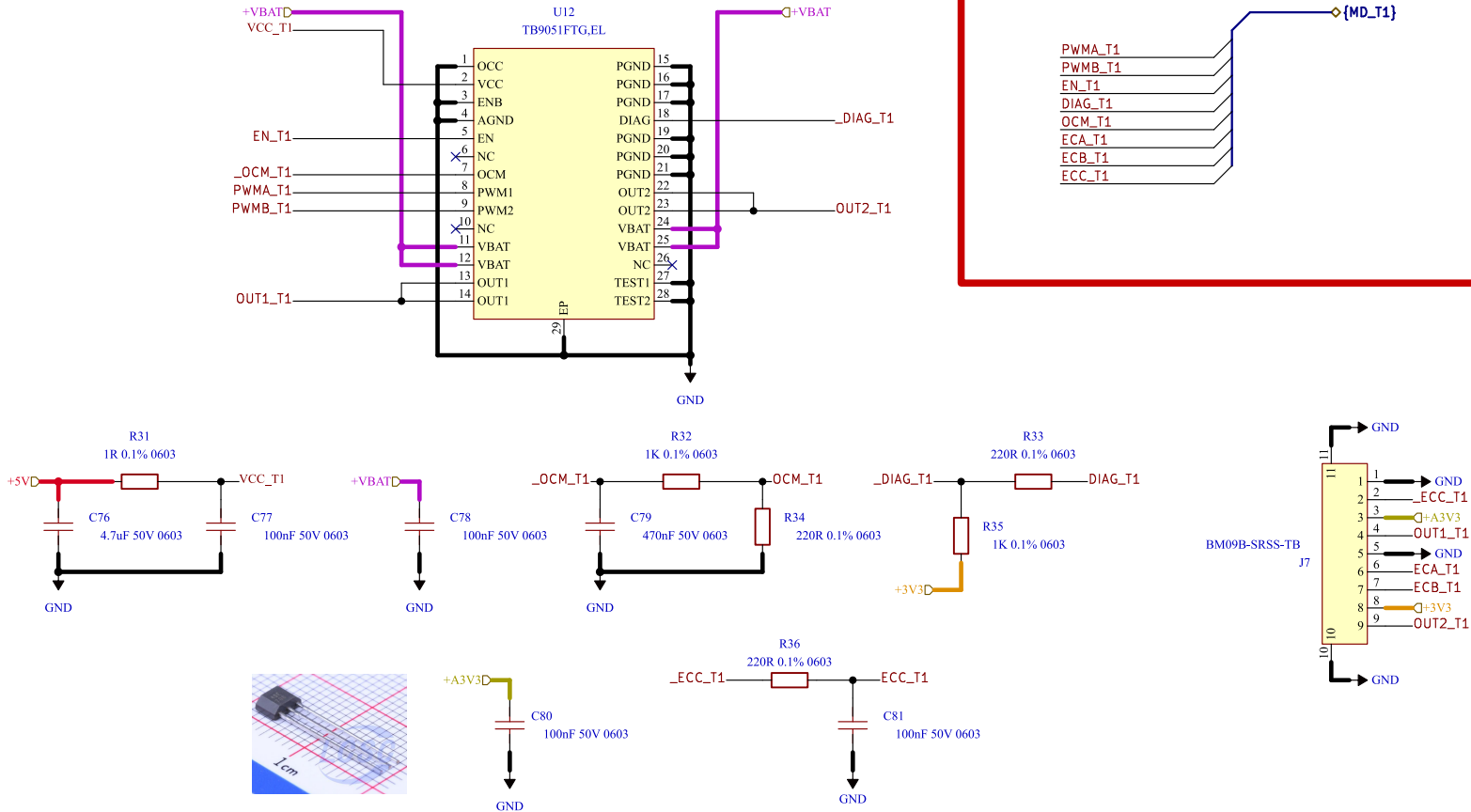
10 of 23

[11] Motor Control - Pinky



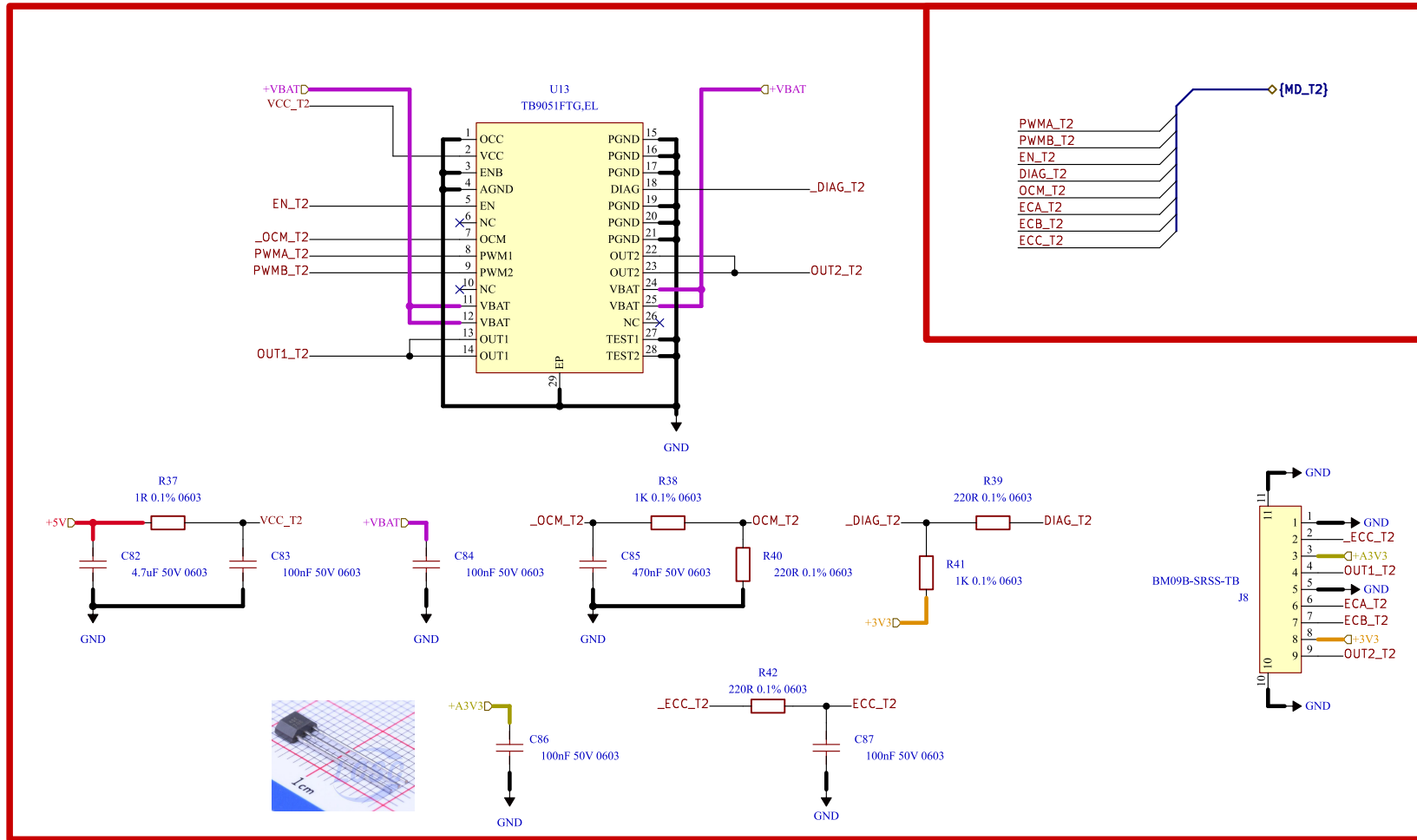
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	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Pinky.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Pinky/		Reviewer:	Size: A4	Sheet: 11 of 23

[12] Motor Control - Thumb 1



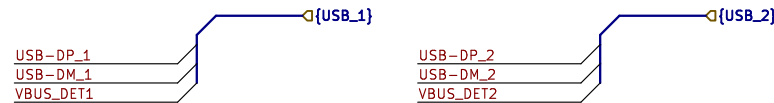
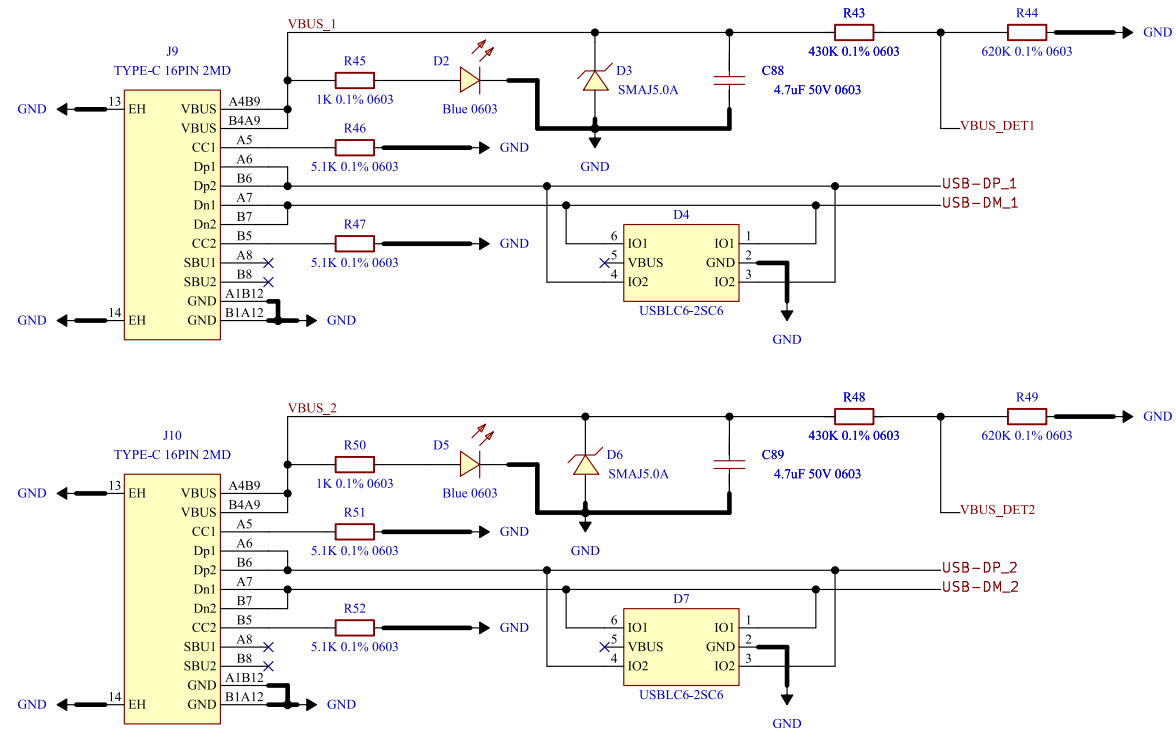
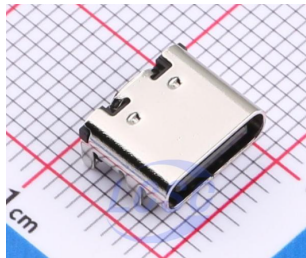
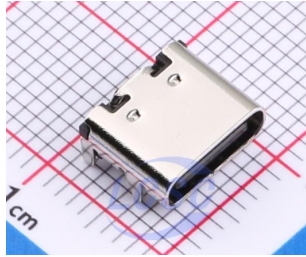
Comments:	Company: Willas		Variant: DRAFT		Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title:	File Name: Motor Control - Thumb 1.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Motor Control - Thumb 1/		Reviewer:	Size: A4	Sheet: 12 of 23	

[13] Motor Control - Thumb 2



Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Thumb 2.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Thumb 2/	Reviewer:	Size: A4	Sheet: 13 of 23	

[14] Interface - USB 2.0



DESIGN NOTE:

- [1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.
- [2] LED direct from VBUS @ 1mA. USB2.0 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.
- [3] VBUS_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.
- [4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPUI 900-1500R).
- [5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HS148+CRS - flashable without crystal.
- [6] Both USB ports fully isolated - no shared nets.
- [7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

538a591

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Interface - USB 2.0.kicad_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Interface - USB 2.0/

Reviewer:

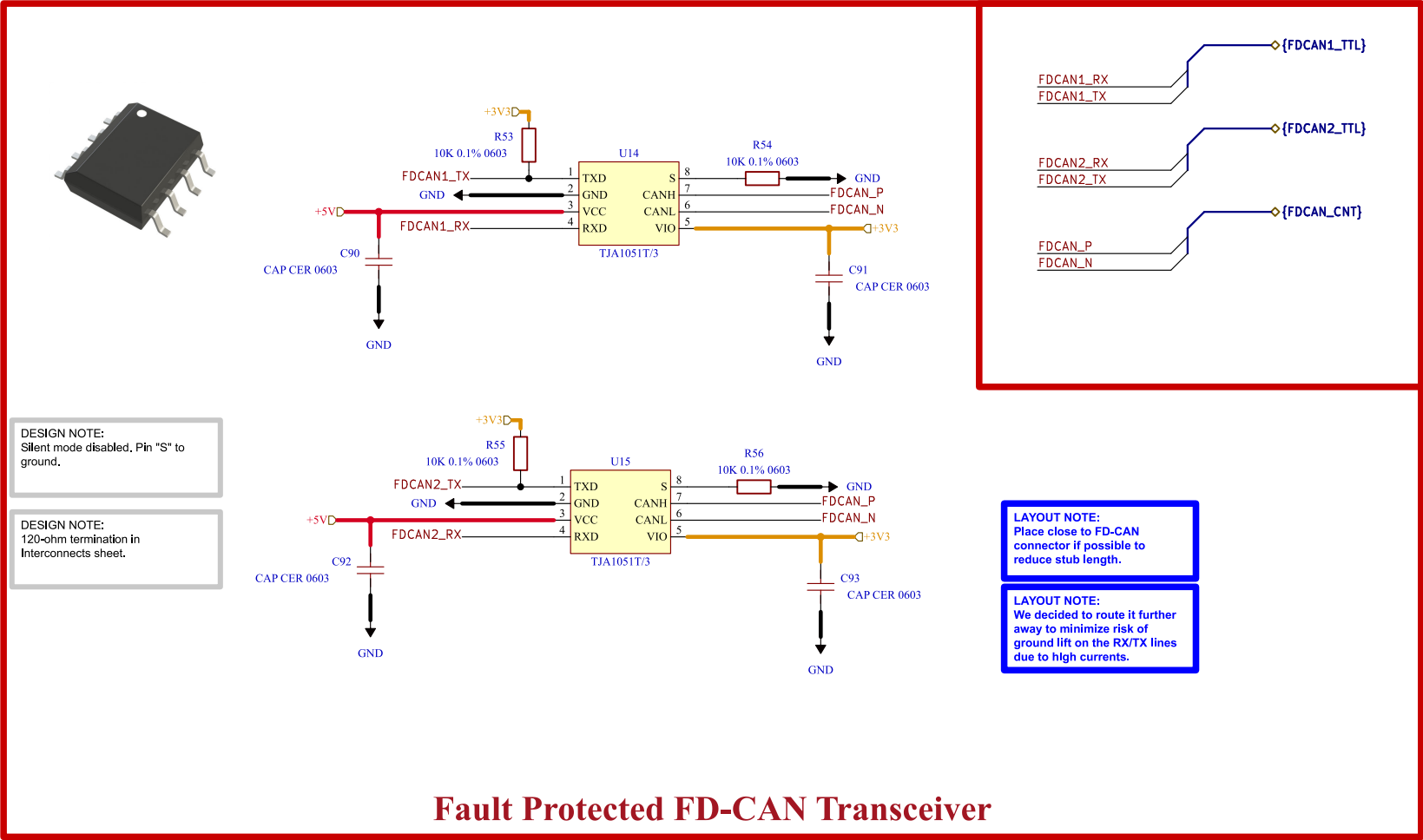
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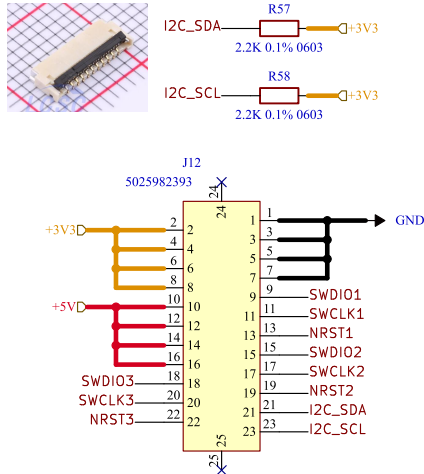
14 of 23

[15] Interface - FD-CAN

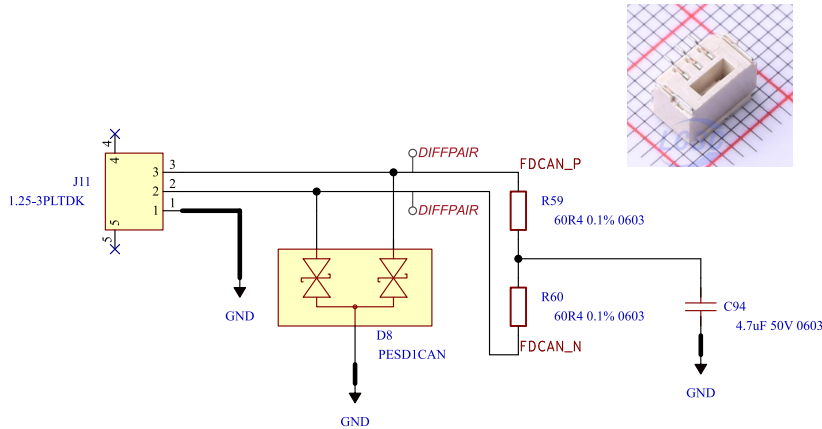


Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title:	File Name: Interface - FD-CAN.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 15 of 23	

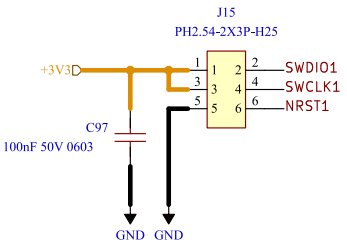
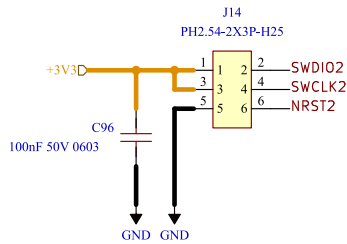
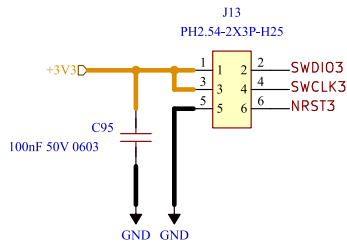
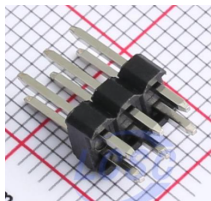
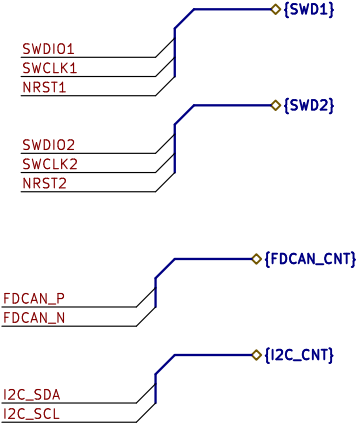
[16] Interface - Interconnects



FFC Connector MCB & MB



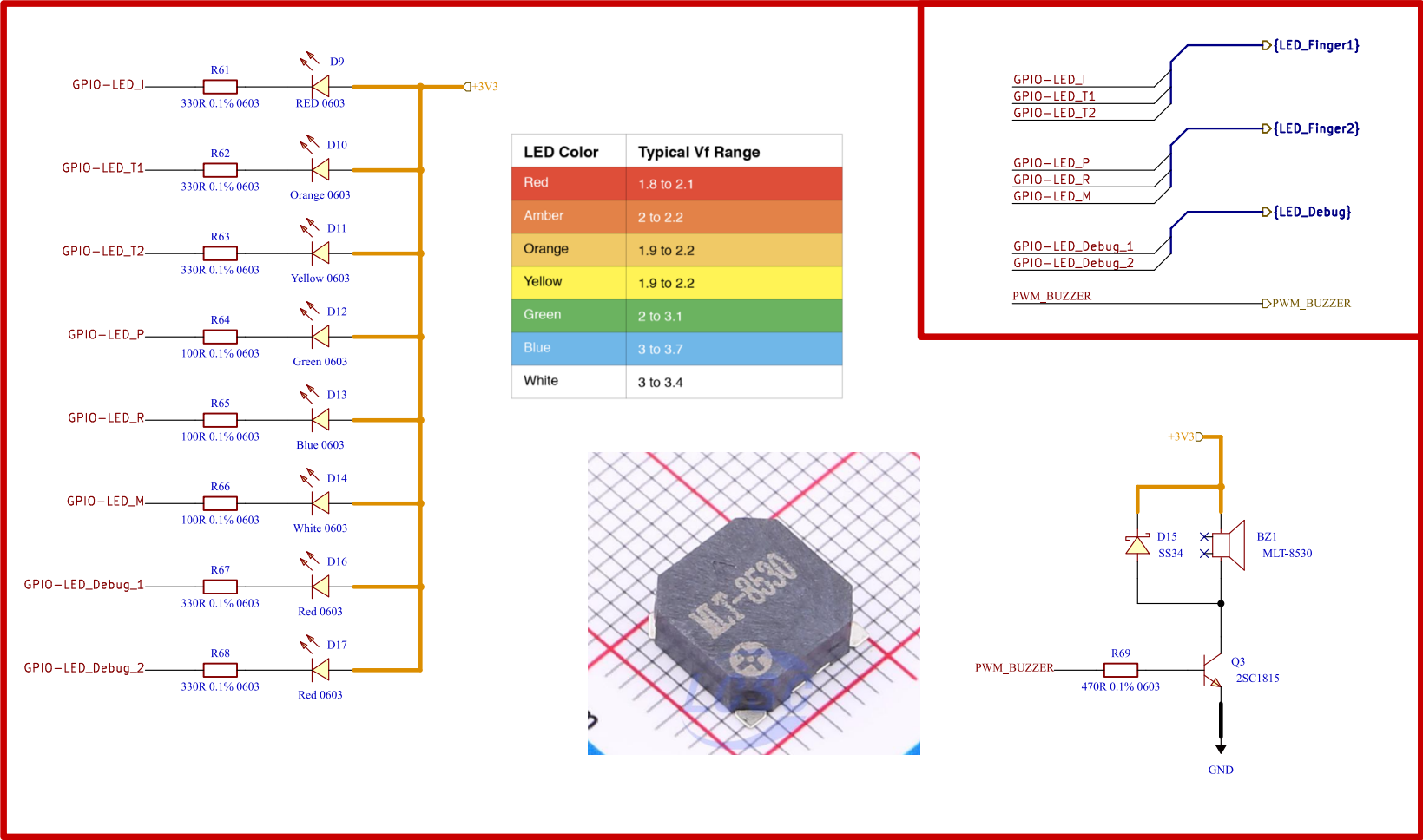
FD-CAN Connector and Termination



SWD Debug Connector

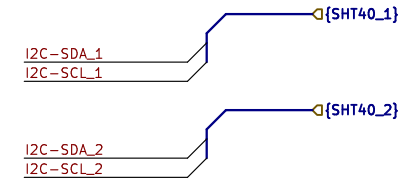
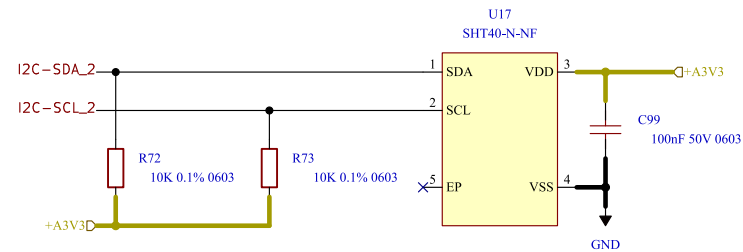
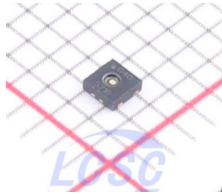
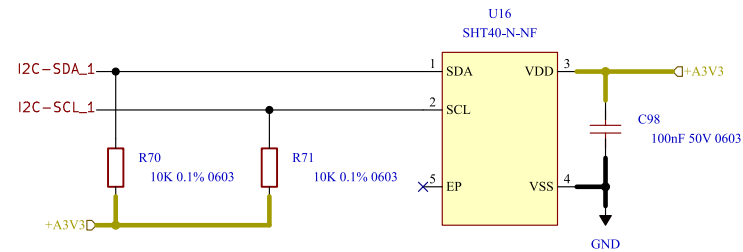
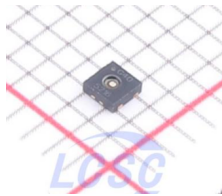
Comments:	Company: Willas 		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Interface - Interconnects.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:	Size: A4	Sheet: 16 of 23

[17] User - LED Indicators & Buzzer



Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: User - LED Indicators & Buzzer.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/User - LED Indicators & Buzzer/		Reviewer:	Size: A4	Sheet: 17 of 23

[18] Sensing - Temperature & Humidity



Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Sensing - Temperature.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Temperature & Humidity/		Reviewer:	Size: A4	Sheet: 18 of 23

[19] Sensing - Battery

DESIGN NOTE:
Battery voltage sensing divider for 2S LiPo battery.

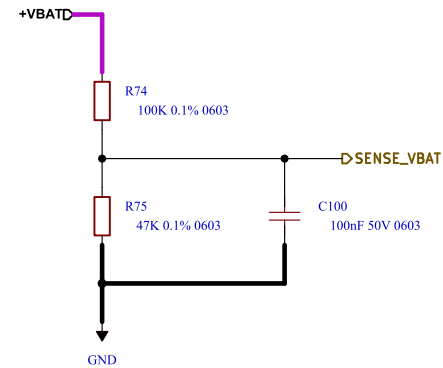
R1 = 100kΩ (Top)
R2 = 47kΩ (Bottom)

Divider Ratio:
 $V_{ADC} = V_{BAT} \times (47 / (100 + 47))$
 $V_{ADC} = V_{BAT} \times 0.3197$

2S LiPo:
Max battery voltage (8.4V): 2.69 V
Nominal voltage (7.4V): 2.37 V
Low battery voltage (6.4V): 2.05 V
Cutoff voltage (6.0V): 1.92 V

Maximum divider current:
 $I = 8.4V / 147k\Omega = 57 \mu A$

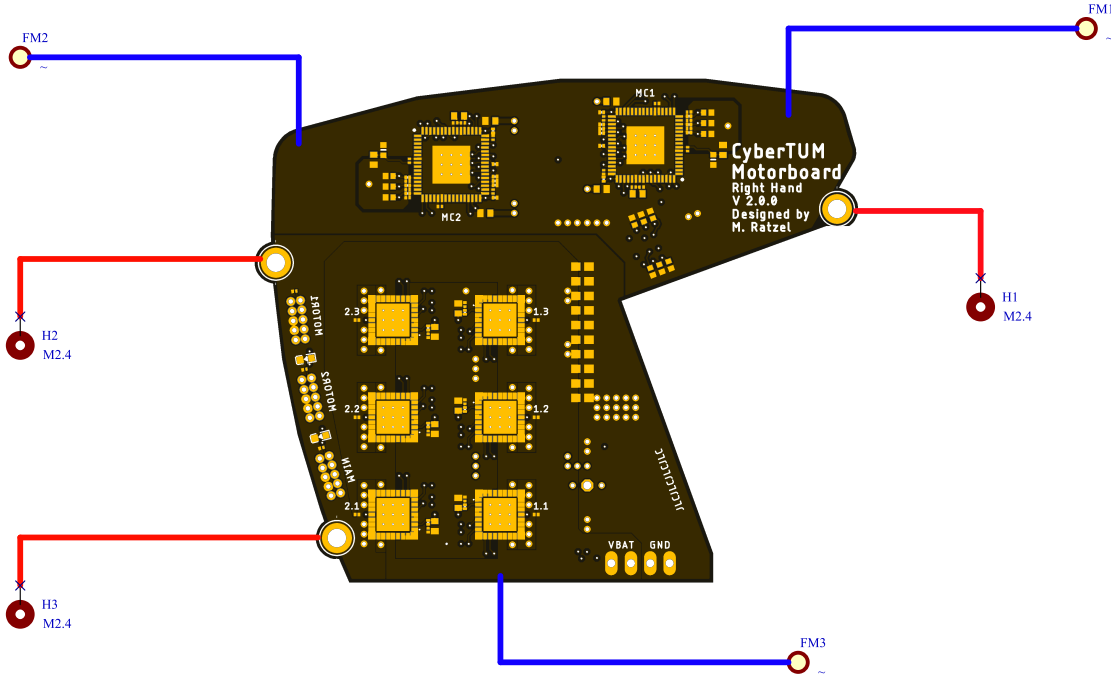
Notes:
- Designed for 3.3V ADC input.
- Add 100nF capacitor at ADC input for noise filtering.
- Use 1% resistors or better.
- Use low temperature coefficient resistors ($\leq 100 \text{ ppm}/^\circ\text{C}$) for improved measurement accuracy.



Battery Voltage Sensing

Comments:	Company: Willas		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Sensing - Battery.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 19 of 23

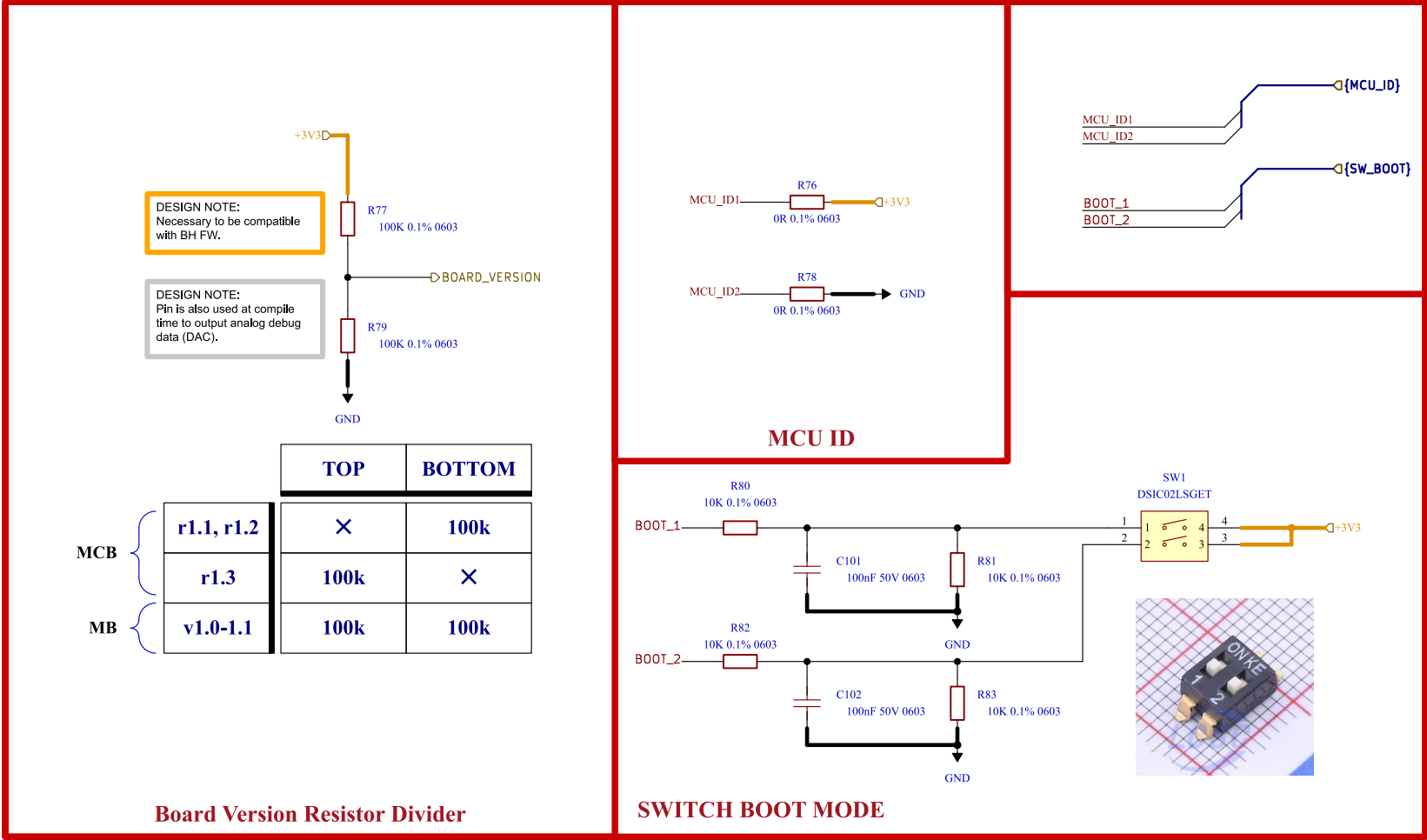
[20] Misc - Holes, Fiducials



Mounting Holes and Fiducial Markers

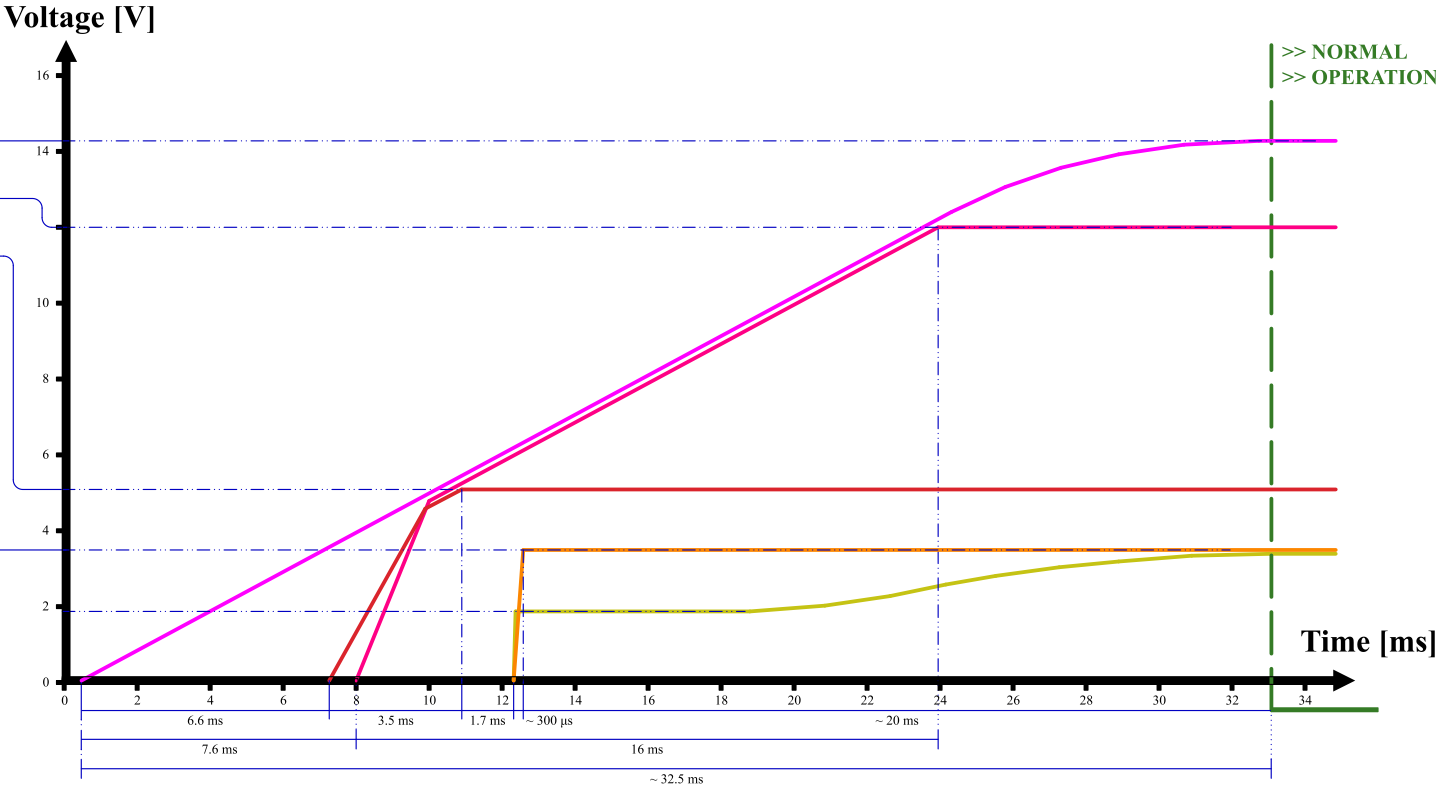
Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 20 of 23

[21] Misc - Board Version, Switch



Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: 538a591	
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Misc - Board Version, Switch.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Misc - Board Version, Switch/		Reviewer:	Size: A4	Sheet: 21 of 23		

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



Comments:	Company: Willas		Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Power - Sequencing.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 22 of 23

Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.	Variant: DRAFT	Git Hash: 538a591
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Revision History.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 23 of 23	